

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	D.HEMASREE	Reg. No.:	192524114
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

CODE:

```
class LoanApp:
```

```
    def __init__(self, id, score, income, debt, amount):
```

```
        self.id = id
```

```
        self.score = score
```

```
        self.dti = debt / income
```

```
        self.amount = amount
```

```
        self.risk = (1 - (score / 850)) + (self.dti * 0.5)
```

```
    def expected_profit(self):
```

```
        return (0.10 * self.amount * (1 - self.risk)) - (0.8 * self.amount * self.risk)
```

```

def process_loans(apps, budget):
    valid_apps = [a for a in apps if a.score >= 600 and a.dti <= 0.5]
    valid_apps.sort(key=lambda x: x.expected_profit(), reverse=True)

    approved = []
    total_spent = 0

    for app in valid_apps:
        profit = app.expected_profit()
        if profit > 0 and (total_spent + app.amount <= budget):
            approved.append(app.id)
            total_spent += app.amount

    return approved

data = [
    LoanApp("User_A", 750, 5000, 500, 10000),
    LoanApp("User_B", 500, 3000, 2000, 5000),
    LoanApp("User_C", 800, 8000, 400, 20000),
    LoanApp("User_D", 650, 4000, 2500, 15000),
    LoanApp("User_E", 700, 6000, 1000, 12000)
]

print("Approved Applicants:", process_loans(data, 35000))
OUTPUT:
Approved Applicants: ['User_C']

```

QUESTIONS

Q1

Banking Loan Approval Risk
Optimization (Decision
Algorithms)
100 marks

Q1 Banking Loan Approval Risk Optimization (Decision Algorithms)

100 marks

A bank must evaluate loan applications to minimize default risk while maximizing profit. Analyze how algorithmic decision-making can be applied. Identify constraints such as credit score, income, and risk factors. Design a solution using optimized decision algorithms. Discuss scalability for large datasets. Evaluate computational complexity. Interpret results in financial risk management.

1/1 answered

Your Code

```
1 class LoanApp:
2     def __init__(self, id, score, income, debt, amount):
3         self.id = id
4         self.score = score
5         self.dti = debt / income
6         self.amount = amount
7         self.risk = (1 - (score / 850)) + (self.dti * 0.5)
8     def expected_profit(self):
9         return (0.10 * self.amount * (1 - self.risk)) - (0.8 * self.am
10 def process_loans(apps, budget):
11     valid_apps = [a for a in apps if a.score >= 600 and a.dti <= 0.5]
12     valid_apps.sort(key=lambda x: x.expected_profit(), reverse=True)
13     approved = []
14     total_spent = 0
15     for app in valid_apps:
16         profit = app.expected_profit()
17         if profit > 0 and (total_spent + app.amount <= budget):
18             approved.append(app.id)
19             total_spent += app.amount
20     return approved
21 data = [
22     LoanApp("User_A", 750, 5000, 500, 10000),
23     LoanApp("User_B", 500, 3000, 2000, 5000),
24     LoanApp("User_C", 800, 8000, 400, 20000),
25     LoanApp("User_D", 650, 4000, 2500, 15000),
26     LoanApp("User_E", 700, 6000, 1000, 12000)
27 ]
28 print("Approved Applicants:", process_loans(data, 35000))
```

Submitted

Input

stdin input

Output

Approved Applicants: ['User_C']

SECTION A – Industry Problem Solving Task

Code of Conduct

I, D.Hemasree/(192524114) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements	Poor understanding; misinterprets problem context

				unclear	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____
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